

# ■ Re-Grow (NatureTech): 4-Minute Hackathon Winning Demo Script

Master Pitch Guide & Live Screen Choreography Aligned to Official Judging Rubric

| Judging Criteria                  | Weight | Re-Grow Winning Strategy & Pitch Focus                                                                 |
|-----------------------------------|--------|--------------------------------------------------------------------------------------------------------|
| Impact & Problem Fit              | 25%    | Translates passive satellite pixels into <b>actionable sapling counts &amp; USD recovery budgets</b> . |
| Technical Execution               | 25%    | FastAPI backend, Next.js 15, OpenCV, rasterio, STAC Sentinel-2, Overpass API & SQLite.                 |
| Usability, Design & Accessibility | 20%    | Glassmorphism dark mode UI, interactive comparison slider, dynamic map, 1-click sample demo.           |
| Creativity & Originality          | 15%    | Active disaster recovery pipeline + 50km geospatial NGO matchmaker + real-time climate telemetry.      |
| Presentation & Demo Quality       | 15%    | Flawless live screen choreography, crisp timing, zero-downtime offline fallback defense.               |

## ■■ MINUTE 0:00 - 0:45 | Impact & Problem Fit (25% Weight)

Rubric Alignment: Demonstrates real-world crisis, target user clarity, and immediate practical utility.

■ Live Screen Action: Open <http://localhost:3000>. Point to the sticky header with API Health Online heartbeat badge and the headline card.

### ■■ WHAT TO SAY (Word-for-Word Spoken Script):

"Honorable judges, every year over 10 million hectares of forest are destroyed by deforestation and wildfires. Today, governments and emergency teams use satellite monitoring to see where trees died. But there is a massive gap: passive maps leave disaster responders completely stranded—with zero actionable budgets, zero sapling counts, and no local ground team to actually plant trees.

Introducing **Re-Grow**—the world's first **Active Disaster Recovery Engine**. Re-Grow transforms raw satellite imagery directly into actionable restoration plans. In under 5 seconds, our platform quantifies precise damage area, calculates required saplings and USD recovery costs, and automatically matches disaster teams with local conservation NGOs in the impact zone!"

## ■■ MINUTE 0:45 - 2:00 | Technical Execution & Usability/Design (45% Weight)

Rubric Alignment: Shows functional Next.js 15 UI, STAC Sentinel-2 live satellite acquisition, and interactive slider UX.

■ Live Screen Action: Click the ■ Live Sentinel-2 Stream tab. Type 'Amazon Rainforest' in the search box, click Locate, then click Fetch Cloud-Free Satellite Data. Show retrieved Sentinel-2 tiles populating the slots.

### ■■ WHAT TO SAY (Word-for-Word Spoken Script):

"Let's see Re-Grow in action. Responders have two options: drag and drop custom high-resolution drone or GeoTIFF files, or click our **Live Sentinel-2 Stream** tab. I'll type 'Amazon Rainforest' and click Locate.

In real-time, our Python FastAPI backend queries Microsoft Planetary Computer STAC API across European Space Agency Sentinel-2 satellites. Notice how it automatically filters for cloud-free tiles and returns baseline historical imagery alongside recent post-disaster satellite imagery.

Now I click **Execute Deforestation Analysis**. Our computer vision engine processes normalized greenness index deltas with adaptive statistical thresholding... and look at this result!"

## ■■ MINUTE 2:00 - 3:00 | Creativity, Originality & Deep Analytics (35% Weight)

**Rubric Alignment:** Highlights original CV algorithms, reforestation economics, live weather telemetry, and NGO matchmaker.

■ **Live Screen Action:** Drag the Interactive Image Comparison Slider left/right. Switch to AI Mask Overlay and click Download Overlay PNG. Point to Live Climate Card and KPI Metrics Cards.

■■ **WHAT TO SAY (Word-for-Word Spoken Script):**  
"Here is our side-by-side satellite image comparison slider. Judges can drag the divider to inspect forest loss pixel-by-pixel, or toggle the **AI Mask Overlay** highlighting detected damage clusters in bright red.

Above our KPIs, Re-Grow pulls real-time environmental telemetry from Open-Meteo—showing temperature, wind speed, and wildfire spread risk.

Now look at the **Reforestation Economics Engine**: Re-Grow detected 6,018 square meters of destroyed tree cover. Applying ecological density standards of 1 tree per 4 square meters, it calculates an exact requirement of **1,504 saplings** and an estimated recovery budget of **\$3,761.25!**"

## ■■ MINUTE 3:00 - 4:00 | Geospatial Matchmaker, CSV Export & Conclusion

**Rubric Alignment:** Shows end-to-end completeness, OpenStreetMap Overpass integration, SQLite audit log, and strong closing.

■ **Live Screen Action:** Scroll to Geospatial OpenStreetMap Map pin, click Export CSV on the Cluster Table, and show the NGO Partner Grid cards.

■■ **WHAT TO SAY (Word-for-Word Spoken Script):**  
"Finally, knowing the budget isn't enough—you need boots on the ground. Re-Grow queries the OpenStreetMap Overpass API to locate registered forestry offices, environmental non-profits, and nature clubs within a 50km radius of the damage.

Responders can click **Connect & Partner** to visit verified websites, export cluster bounding box data via **Export CSV**, or view historical runs logged in our SQLite database.

Re-Grow bridges satellite pixels to real-world saplings planted. Thank you, and we welcome your questions!"

## ■■ Anticipated Judge Q&A; Defense (100% Prepared Answers)

| Judge Question                                                                         | Winning Prepared Answer                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1: What if the satellite STAC API is offline or slow during a live disaster response? | Re-Grow has a zero-downtime architecture. It includes an in-memory & SQLite cache, an offline fallback dataset of top conservation NGOs, and allows users to upload local GeoTIFF/drone images directly. |
| Q2: How accurate is your computer vision damage detection model?                       | We combine normalized greenness index deltas with adaptive statistical thresholding ( $\mu + 1.0\sigma$ ) and morphological opening/closing to filter out shadows and cloud interference.                |
| Q3: How is the \$2.50 per tree cost benchmark calculated?                              | It is modeled on global forestry restoration standards: \$1.00 for sapling purchasing, \$0.75 for land prep and planting labor, and \$0.75 for 2-year survival monitoring.                               |